

Fourth Down Engine: Win-Probability Decisions for Go, Field Goal or Punt

Version: 1.2.0 — 2026-10-08 Author: Will Hooper Data: nflverse play-by-play, 2014–2025 (fit) and 2014–2026 (coach grading)

Abstract

On fourth down an NFL team chooses between going for it, kicking a field goal and punting. This paper describes an engine that scores each option by the team's probability of winning the game afterwards. A gradient-boosted win-probability (WP) model, trained on every regulation snap from 2014 to 2025, values the situations each option can lead to. Sub-models for conversion, field-goal success and punt outcome supply the odds of each situation. The clock, timeouts, the two-minute warning, the second-half kickoff, the pre-game line, weather, roof, altitude and the kicker all enter the calculation, and each optional factor can be switched off. A whole-pipeline bootstrap measures how sure each call is. The engine then grades every current head coach by replaying each of their fourth downs twice, once as called and once as the engine would call it, and charging only the calls the bootstrap is confident were wrong. It also grades every kicker against a measured replacement level. All results are in section 7; every figure there is rendered from the pipeline's output files.

1. The question

A fourth-down call is a choice between lotteries. Going for it risks giving the opponent a short field for the chance of keeping the drive. A field goal trades a likely 3 points for possession. A punt gives up the ball for field position. Which lottery is best depends on the whole game state: a 3-point lead is worth little in the first quarter and almost everything with a minute left.

Expected points (Carter & Machol 1971; Romer 2006) answer the question early in games, when the score barely matters. Win probability answers it at every point, because it already accounts for score and clock (Burke; Lock & Nettleton 2014; Yurko, Ventura & Horowitz 2019; Baldwin 2021). This engine uses win probability throughout.

The same machinery answers a narrower early-down question: on 1st, 2nd or 3rd down with the clock running out, should the team kick the field goal now instead of running another play (section 3.3)?

Scope. Regulation only. Overtime plays are excluded from training, and a game tied at the end of regulation counts as 0.5. Two-point decisions, fake kicks as a separate option, and onside kicks are out of scope.

2. Data

nflverse play-by-play (ADR-0001): one parquet file per season, the same source as nflfastR and nfl4th. The fit uses 2014–2025; the coach grader also reads 2026 (weeks played so far), which the model has never seen.

Derived columns, all from the offense's point of view:

- `score_diff`: offense score minus defense score.
- `spread`: points the offense was favored by before the game (nflverse `spread_line` is from the home side; it is negated for the away team).
- `receive_2h`: 1 in the first half if the offense will receive the second-half kickoff, derived from the first kickoff of the game.
- `tmw_pending`: 1 while more than 120 seconds remain in the half, i.e. the two-minute warning is still to come.
- Weather: wind and temperature from the structured fields, filled from the free-text weather string where missing (46% of outdoor games in 2022 lack the structured fields). Precipitation is a keyword match on the same string. Indoors, wind and precipitation are 0 and temperature is 70 °F.
- Altitude: 5,280 ft for Denver home games, 7,350 ft for Estadio Azteca.

3. Method

3.1 Win-probability model

For a snap with features $\mathbf{x}$, the model gives the offense's probability of winning:

$$WP(\mathbf{x}) = \sigma(b_0 + \sum_t f_t(\mathbf{x})), \sigma(z) = 1 / (1 + e^{(-z)})$$

where each f_t is a regression tree (250 trees, ≤ 31 leaves, gradient boosting on log loss; scikit-learn `HistGradientBoostingClassifier`). Inputs: score difference, game seconds, half seconds, half, yard line, down, distance, both teams' timeouts, two-minute warning pending, second-half receiver, spread, home, and two engineered terms from nflfastR (Baldwin 2021):

$$\text{spread_time} = \text{spread} \cdot e^{(-4 \cdot s)}, \text{diff_time_ratio} = \text{score_diff} / e^{(-4 \cdot s)}$$

with s the elapsed share of the game. The first lets the pre-game line fade as the game reveals itself; the second lets a lead grow in value as time runs out. Monotone constraints force WP to rise with score difference, field position, own timeouts, spread and both engineered terms, and to fall with the opponent's timeouts. The trees are exported to JSON and evaluated identically in Python and JavaScript (ADR-0002). Ties are dropped from training.

3.2 Option values

Let s be the fourth-down state and $V(s')$ the WP of a next state s' from **our** point of view ($V = WP$ if we have the ball, $1 - WP$ if they do). Then

$$\begin{aligned} \text{Go} &= p_c \cdot E[V(s' \mid \text{convert})] + (1 - p_c) \cdot V(s' \mid \text{stopped}) \\ \text{FG} &= p_m \cdot V(s' \mid \text{made}) + (1 - p_m) \cdot V(s' \mid \text{missed}) \\ \text{Punt} &= (1 - p_{td} - p_{mf}) \cdot E[V(s' \mid \text{normal punt})] + p_{td} \cdot V(s' \mid \text{return TD}) + p_{mf} \cdot V(s' \mid \text{muff}) \end{aligned}$$

and the call is the option with the largest value (Romer 2006; nfl4th). The next states are built as follows.

- **Converted.** Yards gained come from the empirical distribution of gains on successful 3rd and 4th downs with similar distance (20 quantiles per distance bucket). A gain that reaches the end zone is a touchdown (+7, then the opponent receives a kickoff). Otherwise it is our 1st & 10 (or & goal).
- **Stopped.** The opponent takes over where the play ended (mean yards gained on failed attempts for that distance).
- **Touchdown.** +6, then the scoring team takes the extra point or the 2-point try, whichever gives it the better win chance (rates measured since 2015, when the extra point moved back). Late in a game this matters: down 8, the engine values a touchdown as the 2-point try that can tie.
- **Made field goal.** +3, then the opponent receives a kickoff.
- **Missed field goal.** The opponent takes over at the spot of the kick (line of scrimmage + 7), or at its own 20 if that is further from its goal. Kicks longer than 70 yards, the longest attempt in the data, are out of range and the option is removed.
- **Punt.** The opponent's start spot comes from the empirical distribution of real punts from the same 5-yard band. A return touchdown (+7 for them, our kickoff return) and a muff recovered by the kicking team are separate outcomes, with rates shrunk toward the league rate (200 pseudo-counts).
- **Fair-catch kick.** After a fair catch, NFL Rule 11-4-3 lets the receiving team try a free-kick field goal from the catch spot: a place kick or a drop kick, with no rush, from (spot + 10) yards. If time expires during the punt, it is an untimed down. When 10 seconds or fewer would remain after the punt (`fair_catch_kick_window`), each fair catch (rate measured per 5-yard band) gives the opponent the better of that kick (league-average kicker, same weather and stadium) and playing on. Good punts leave the receiver 70+ yards from the goal posts, so the threat is real mainly after short punts at the end of a half.
- **Drop kicks from scrimmage** are legal on any down and score the same 3 points, but they are almost never used and leave no data for a separate make rate. The engine treats a drop-kicked field goal as a field goal.
- **Kickoffs** place the receiver at the mean start spot. The app uses the current rules (`rules_season`). The coach grader uses each season's own spot, measured from that season: the touchback moved from the 20 to the 25 in 2016, the 30 in 2024 and the 35 in 2025, and receivers started near their own 22, 25 and 31 in those eras. (Before 1.2.0 every season was graded with the 2025 spot, which undervalued scores in older seasons.)

Clock. Each outcome uses the median game-clock time from this snap to the next snap, measured from the data separately for the final two minutes of a half. If the two-minute warning is pending and the play would cross 2:00, the next state starts at 2:00 and the warning is spent. If the half ends, a first half resets to the second-half kickoff (the receiver gets the ball, timeouts reset); a second half ends the game, with WP 1 or 0 by the final score.

Overtime. Since 2022 (playoffs) and 2025 (regular season) both teams get a possession in overtime, so the coin toss decides much less than it did. A tie at the end of regulation is valued as

$$V_{\text{tie}} = 0.5 + (1 - p_{\text{tie}}) \cdot (\sigma(\beta \cdot \text{spread} / 10) - 0.5)$$

where β is fit on every decided overtime game in the data (the better team's edge) and p_{tie} is the measured share of overtime games that end tied (half a win). With the team-strength factor off, $V_{\text{tie}} = 0.5$. Only about twenty games so far fall under the both-possess rules, too few to fit separately. Timeouts carry over unchanged.

3.3 Early downs: kick now or run a play?

On 1st–3rd down the options are **run a play** and **kick a field goal now** (nobody punts). Running a play keeps the kick available on a later down, so it wins almost always. The exception is the clock:

$$\text{Play} = (1 - p_{\text{to}}) \cdot E_g[V(s' \mid \text{gain } g)] + p_{\text{to}} \cdot V(s' \mid \text{turnover})$$

Gains g come from the empirical distribution of 1st-, 2nd- or 3rd-down runs and passes at similar distance (20 quantiles); p_{to} is the measured turnover rate for that down. A gain past the line is a new 1st down; otherwise it is the next down at the new distance. The clock runs by the measured snap-to-snap time, separately for plays that gain a first down (clock keeps running) and plays that do not (incompletions stop it). If that time would end the half and the offense has a timeout, it calls one, and the play costs only the measured length of a play followed by a timeout. With no timeout left it spikes the ball, which costs a down (allowed only while it leaves at least 4th down) and the measured time from the play's snap through the spike to the next snap. If neither saves the clock, the half ends and a team that is behind loses the chance to kick. The WP model values every next state.

3.4 Conversion

$$\text{logit } p_{\text{c}} = \beta_0 + \beta_1 \cdot \text{is4} + \beta_2 \cdot \min(d, 20)/10 + \beta_3 \cdot \ln d + \beta_4 \cdot \text{goal} + \beta_5 \cdot \text{inside10} + \beta_6 \cdot \text{redzone} [+ \beta_7 \cdot \text{spread}/10]$$

fitted by maximum likelihood on all 3rd- and 4th-down runs and passes, with `is4 = 1` at prediction time. Pooling 3rd downs follows Romer (2006), who found the 3rd/4th difference small, and Brill et al. (2025), whose best conversion model pools them. A defensive penalty that gives a first down counts as a conversion, as in nfl4th. Terms that let 4th-down rates differ by distance were tested in 1.1.0 and left out: they moved hold-out log loss by 0.0005.

4th & 1 precision. Play-by-play records yards to go as a whole number, so a recorded "4th & 1" runs from inches to more than a yard. With tracking data, Lopez (2020) found teams that went were 0.70 yd from the line and teams that kicked 0.98 yd. The engine treats a recorded 1 as the average true distance (0.85 yd) and shifts the distance the conversion model sees by ($\text{true} - 0.85$). In the app the user can say "inches" (0.35 yd) or "about a yard" (1.0 yd). The coach grader judges a team that went at 0.70 yd and a team that kicked at 0.98 yd, so a coach is not charged for passing on a 4th & 1 that was really a full yard. The spread term is used when the team-strength factor is on.

3.5 Field goals

$$\text{logit } p_{\text{m}} = \alpha_0 + \alpha_1 \cdot D + \alpha_2 \cdot D^2 + \alpha_3 \cdot \text{era} [+ \text{weather terms}] [+ \text{stadium terms}] [+ a_{\text{k}} + b_{\text{k}} \cdot (D - 4)]$$

with D the kick distance in tens of yards. Weather terms: wind, wind $\times$ distance, cold (degrees below 50 °F), precipitation. Stadium terms: dome or closed roof, altitude. These follow Clark, Johnson & Stimpson (2013)

and Pasteur & Cunningham-Rhoads (2014). The era term (linear year trend, a 2020+ step, or both) is chosen on the hold-out seasons. Four models are fit, one for each on/off combination of the weather and stadium factors, so an off factor is marginalised rather than set to a guess. a_k and b_k are the kicker's accuracy and range (section 4).

3.6 Factors as toggles

Each optional factor (weather, stadium, team strength, kicker) can be switched off. Off means: the matching field-goal and punt models without that factor's terms; a spread of 0 (an even matchup) in the WP and conversion models; a league-average kicker. The core inputs (score, clock, field position, distance, timeouts, two-minute warning, second-half receiver, home) are always on.

4. Kickers

Skill model. For kicker k , relative to the environment-adjusted league logit z :

$$\text{logit } P(\text{make}) = z + a_k + b_k \cdot (\text{yards} - 40) / 10$$

a_k (accuracy) moves the whole curve; b_k (range) tilts it, so a big leg holds its make rate at long distance. This matches the kicker intercepts and distance slopes of Long (2019) and the distance-varying shrinkage of Osborne & Levine (2017). Each (a_k, b_k) is a maximum a posteriori estimate under independent normal priors $N(0, \sigma_a^2)$ and $N(0, \sigma_b^2)$, found by Newton's method. The prior widths are chosen on a grid by hold-out log loss. Skill is measured against the fullest environment model, so a Denver kicker is not credited for altitude.

Replacement level. Following nflWAR (Yurko et al. 2019): in each season, kickers are ranked by attempts, and all kicks by anyone outside the top 32 are pooled into one unshrunk (a_R, b_R) . A second rule (short careers inside the data) is reported as a sensitivity check.

Value. For a kicker's actual kicks i ,

$$\text{points over replacement} = 3 \cdot \sum_i [\sigma(z_i + a_k + b_k x_i) - \sigma(z_i + a_R + b_R x_i)]$$

reported in total and per 100 kicks. Grades rank kickers active in the latest season by points over replacement per 100 kicks.

5. Grading coaches: every call played twice

For each fourth-down run, pass, punt or field goal by a team (penalties, kneels and spikes excluded), the engine evaluates the same state with every factor on, using that game's kicker, weather, roof and line:

- **Coach's way:** V_c = value of the option the coach chose.
- **Engine's way:** V_e = value of the best option.
- **Cost:** $V_e - V_c \geq 0$, the expected wins the call gave up.
- **What happened:** V_r = WP at the next snap after the real play (or the final result if the game ended), from the coach's side.

- **Luck:** $V_r - V_c$. Luck sums to zero in expectation; it measures what the dice did after the decision, not the decision.

Confidence. The whole pipeline is refit on 20 bootstrap resamples of whole games (section 6). A call is a **confident mistake** only when at least 90% of replicates rank another option above the coach's choice. Following Brill, Yurko & Wyner (2025), the graded figure sums cost over confident mistakes only; the sum over all disagreements is reported alongside (ADR-0003).

Grades. A coach's figure is confident-mistake cost per game, expressed as wins per 17 games, with a 90% interval from resampling his calls. Letters rank him against every head coach with at least 17 games in the data: the best 10% earn A+, then A (to 25%), B+ (40%), B (60%), C+ (75%), C (90%) and D. Coaches with fewer than 17 games are listed as Incomplete.

This week. During the season the same grading runs every Tuesday on the current season's games (engine/weekly.py, scheduled on GitHub). The app lists each week's costliest confident mistakes and its best gutsy calls: the coach went for it, the engine agreed by 3 or more points, and 90% of replicates agree.

6. Uncertainty and validation

Bootstrap. Twenty replicates, each a full refit (WP, conversion, field goal, punt, clock) on games drawn with replacement. Games are the unit because plays in a game share one outcome. Brill et al. also resample drives within games; this engine does not, which may understate variance (risk 4). Confidence moves in steps of 5%.

Validation. Every model is fit on 2014–2023 and scored on 2024–2025. Splits are by season, so no game is on both sides. The key test is not WP accuracy alone but whether **option values** are calibrated: for the option a coach actually chose, does the engine's predicted WP match how often the team won?

Edge calibration. Option values can be calibrated on average while the differences between options are not. So the graded decisions are also binned by the engine's predicted edge for going, and within each bin teams that went are compared with teams that kicked: predicted gap against realized gap (engine/edge_check.py). Teams that went often knew something the engine cannot see, so the realized gap is biased toward going; the ratio is an upper bound on how much of the predicted edge is real.

What validation does not prove. That following the engine wins more games (no randomized test of 4th-down policy exists); that any single call is right; or that short-yardage conversion is unbiased (section 8).

7. Results

Headline results

- Win-probability log loss on held-out 2024, 2025: **0.446** (nflfastR `vegas_wp` on the same 81,654 plays: 0.4457).
- Option values on held-out 4th downs, predicted vs actual win rate: go 37.3% vs 38.1% (n = 670); field goal 56.4% vs 55.6% (n = 826); punt 47.2% vs 46.6% (n = 1501).

- League cost of 4th-down calls, confident mistakes only: **0.84** wins per team-season in 2014, **0.62** in 2025. Counting every disagreement: 1.11 and 0.86.
- Coaches went for it on 12.7% of 4th downs in 2014 and 24.0% in 2025; the engine clearly favored going (edge of 1 point or more) on 30.2% and 34.5%.
- Share of 4th downs that are toss-ups (edge under 1 WP point), 2014–2026: 37.9% to 45.2%. Share of calls the bootstrap is confident about: 47.7% to 49.9%.
- Current head coaches with the lowest confident-mistake cost: Liam Coen 0.34 wins/17 games (A+); Sean McDermott 0.46 wins/17 games (A+); Matt LaFleur 0.46 wins/17 games (A+).
- Highest: Shane Steichen 0.86 (C); Todd Bowles 0.83 (C+); Sean Payton 0.81 (C+).
- Edge check: relative to teams that kicked, teams that went did -0.5 ± 0.6 WP points versus the engine's prediction (negative = the engine overstates going; selection flatters going, so near zero does not clear it).
- Replacement-level kicker: 76.4% made vs 85.5% expected; 61% from 50 yd vs 77% league average.
- Top current kickers by points over replacement per 100 kicks: B.Aubrey 44.2; C.Boswell 40.8; W.Reichard 39.5.

Full tables: `docs/reports/validation-report.md`, `docs/reports/coach-grades.md`, `docs/reports/kicker-grades.md`.

Benchmark against the literature. Published estimates of the cost of fourth-down conservatism are about 0.4 wins per team-season before 2016 (Romer 2006; Yam & Lopez 2019), 0.35 for 2017–2019, and 0.22 after correcting for exact distance (Lopez 2020). The engine's toss-up share and confident share agree with Brill, Yurko & Wyner (2025), who found 44% of edges under 1 point and 48% of decisions confident. Its share of clear "go" calls is in the range of published fourth-down bots. **Its league cost totals above are larger than every published estimate, roughly two to four times the distance-corrected figure.** The most likely causes are the ones in section 8: whole-number distance (Lopez found it inflates the apparent gain by about 40%), the separate gains and conversion models, and a bootstrap that measures sampling noise but not model error. Read the coach figures as a **ranking** of how closely each coach's calls match the engine, not as a precise count of wins lost. The letter grades are percentile ranks for that reason.

8. Limitations

1. **Distance is a whole number.** Teams that go on "4th & 1" are closer to the line than teams that kick (Lopez 2020). The engine cannot see this, so its short-yardage "go" edge is probably biased upward. Lopez found the bias was about 40% of the apparent benefit of aggression.
2. **Long-yardage attempts are selected.** Hold-out conversion on 4th & 11+ is higher than predicted, because those tries happen mostly in desperate late-game spots against soft defenses. The engine is, if anything, conservative on long tries.
3. **Gains and conversion are modelled separately.** nfl4th fits one joint yards-gained distribution that depends on field position and team strength. This engine bins gains by distance only.
4. **One kickoff spot.** Kickoff returns are a mean, not a distribution.
5. **Touchdown = 7.** The extra point is treated as certain; two-point decisions are not modelled.
6. **Grading uses in-sample models for 2014–2025.** Option values validate out of sample, and 2026 calls are fully out of sample.

7. **Field-goal link.** Osborne & Levine (2017) found a complementary log-log link fits better than logit. It was not tested here.

9. Relation to the literature

Choice	This engine	Literature
Decision rule	Max WP over probability-weighted outcomes	Romer (2006); nfl4th; Brill et al. (2025)
WP model	Boosted trees, monotone, spread-fade and score/time terms	nflfastR XGBoost (Baldwin 2021); GAM in nflWAR
Conversion	Logistic, 3rd + 4th downs pooled, 4th-down term, defensive-penalty firsts	Romer (2006); nfl4th (joint yards-gained XGBoost)
Field goal	Logistic with environment terms and kicker accuracy/range	Clark et al. (2013); Pasteur & Cunningham-Rhoads (2014); Osborne & Levine (2017); nfl4th uses no kicker or weather
Punt	Empirical quantiles by 5-yd band, return TD and muff rates	nfl4th kernel density
Uncertainty	20-replicate game bootstrap; grade confident mistakes only	Brill, Yurko & Wyner (2025)
Replacement kicker	Outside top 32 by attempts per season	nflWAR (Yurko et al. 2019)

Coaches' conservatism is consistent with risk aversion over the next state's value (Sandholtz et al. 2024).

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